

SHIRAKAWA, Japan

WMO: 475970

Lat: 37.1321N

Lon: 140.2150E

Elev: 1170

StdP: 14.09

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 21.6	(c) 23.7	(d) 10.3	(e) 9.7	(f) 31.0	(g) 12.2	(h) 10.7	(i) 31.1	(j) 28.6	(k) 34.1	(l) 26.1	(m) 34.8	(n) 4.9	(o) 290	(p) 0.474

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 14.1	(c) 89.0	(d) 75.4	(e) 86.6	(f) 74.6	(g) 84.0	(h) 73.3	(i) 77.1	(j) 85.4	(k) 76.0	(l) 83.4	(m) 74.9	(n) 81.4	(o) 8.8	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
74.8	136.3	80.4	73.8	131.6	79.6	72.8	127.3	78.7	41.5	85.6	40.4	83.8	39.3	81.5	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 25.2	(b) 21.7	(c) 19.1	(e) 17.2	(f) 92.7	(g) 1.9	(h) 1.8	(i) 15.8	(j) 94.0	(k) 14.6	(l) 95.0	(m) 13.6	(n) 96.0	(o) 12.2	(p) 97.3		
			DB	WB	WB	WB	WB	WB	WB	WB	WB	WB	WB	WB		
			17.2	92.7	1.9	1.8	15.8	94.0	14.6	95.0	13.6	96.0	12.2	97.3		
			15.9	78.8	1.9	1.1	14.5	79.6	13.4	80.2	12.4	80.9	11.0	81.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	54.0	33.2	34.3	40.3	50.5	60.3	66.9	74.1	76.0	68.6	57.8	46.7	37.8
	DBStd	15.79	4.18	4.88	5.83	6.55	5.34	4.86	5.31	4.97	5.87	5.46	5.48	4.94
	HDD50	1855	522	439	307	73	3	0	0	0	0	5	128	378
	HDD65	4900	987	858	766	436	161	35	4	2	30	231	548	842
	CDD50	3307	0	0	6	87	321	506	746	804	557	248	31	1
	CDD65	875	0	0	0	0	15	91	284	341	136	8	0	0
	CDH74	5525	0	0	0	14	148	443	1864	2384	650	22	0	0
	CDH80	1612	0	0	0	1	19	63	607	797	125	0	0	0
(14) Wind	WSAvg	7.9	8.6	9.2	9.9	9.6	8.5	6.7	6.2	6.0	6.6	7.2	7.7	8.1
(15) Precipitation	PrecAvg	56.6	2.1	1.8	3.0	4.1	4.6	5.6	8.5	7.9	7.9	5.7	3.2	2.2
	PrecMax	78.3	5.2	4.4	5.0	7.8	9.4	9.7	17.0	28.1	14.7	18.7	8.0	5.4
	PrecMin	40.6	0.7	0.4	0.8	0.6	1.9	2.4	2.5	1.8	0.4	0.7	0.4	0.6
	PrecStd	9.0	1.2	0.9	1.0	2.0	1.9	1.9	3.2	5.1	3.4	4.1	1.8	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	51.1	56.7	65.7	75.8	82.9	85.1	92.0	92.5	86.8	77.2	66.6	57.2
		MCWB	42.6	46.3	50.1	56.8	63.7	69.5	75.4	76.6	73.6	66.1	56.3	49.3
	2%	DB	46.7	50.8	59.8	70.5	78.1	81.7	88.9	89.6	83.7	72.4	62.6	52.8
		MCWB	39.0	41.9	47.2	54.4	61.3	68.1	75.2	76.0	72.4	63.2	54.2	45.8
	5%	DB	43.6	47.0	55.4	66.7	74.6	79.0	86.4	87.3	80.6	69.6	59.7	49.7
		MCWB	36.5	39.1	45.9	53.6	59.8	66.7	74.5	75.3	71.0	61.8	52.3	43.2
	10%	DB	41.1	43.6	51.7	62.9	71.4	76.2	83.5	84.8	77.7	67.1	57.0	46.9
		MCWB	34.8	36.5	43.4	51.5	58.8	65.8	73.3	74.7	70.1	59.9	51.2	41.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.3	48.5	53.8	62.6	67.4	73.6	78.2	78.9	75.8	69.0	59.2	51.6
		MCDB	50.7	55.1	59.1	67.9	75.1	79.4	87.0	87.5	82.4	73.6	63.2	54.0
	2%	WB	39.9	43.0	50.4	58.3	64.7	71.5	76.8	77.5	74.4	66.0	56.4	47.2
		MCDB	45.4	48.6	57.2	66.5	72.5	77.1	85.2	85.8	80.1	70.0	59.9	51.2
	5%	WB	37.4	39.8	47.2	55.7	62.8	69.8	75.6	76.6	73.1	63.6	54.3	44.1
		MCDB	42.5	45.8	54.3	62.0	69.9	75.6	83.1	84.3	78.5	67.6	58.1	49.0
	10%	WB	35.2	37.0	44.1	53.5	61.0	67.9	74.5	75.5	71.6	61.3	51.8	41.4
		MCDB	40.2	42.9	50.8	60.0	68.0	73.5	81.1	82.4	76.3	65.8	56.3	46.3
(35) Mean Daily Temperature Range	5% DB	MDBR	13.5	14.6	16.8	18.5	17.8	14.9	13.6	14.1	13.8	15.1	16.2	14.4
		MCDBR	18.0	21.0	24.1	25.0	24.2	19.9	18.5	17.8	16.4	17.2	20.2	18.7
		MCWBR	11.8	13.8	14.9	12.9	11.0	8.8	7.2	6.8	7.4	9.6	12.9	12.8
	5% WB	MCDBR	17.3	19.9	21.3	18.7	18.4	14.8	16.1	16.0	14.2	14.1	16.8	17.8
		MCWBR	12.2	14.2	15.2	12.0	10.5	7.6	7.0	6.9	7.0	9.3	12.2	13.3
(40) Clear-Sky Solar Irradiance	taub		0.353	0.398	0.468	0.519	0.540	0.570	0.573	0.568	0.489	0.436	0.386	0.354
	taud		2.217	2.076	1.935	1.872	1.871	1.865	1.920	1.927	2.102	2.170	2.237	2.269
	Ebn at Noon		258	258	250	245	242	234	233	230	242	243	243	248
	Edh at Noon		34	44	56	62	64	64	60	59	47	40	33	30
(44) All-Sky Solar Radiation	RadAvg		719	941	1222	1467	1630	1513	1452	1453	1177	942	725	626
	RadStd		45	72	75	135	164	108	227	169	96	90	54	43

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	+0.88	+1.13	N/A	N/A	N/A
(47) Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page